Master's Thesis

Microlocal Morse Theory and Truncated Microsupport
（超局所モース理論と打切り超局所台）

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1 Introduction

1.1 Microlocal Sheaf Theory

Microlocal Morse theory is a machinary to study the propagations of a cohomology of (complexes of) sheaves. More generally, this theory is considered as a part of microlocal sheaf theory, due to Kashiwara and Schapira. They defined the notion of microsupport of (a complex of) sheaves around 1980's.

To explain more precisely, Let fix some notations. We consider a commutative unital ring $\mathbf{k}$ of finite global dimension (e.g. $\mathbf{Z}$ or a field). Let X be a topological space. We denote by $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ (resp. $\mathsf{D}^{+}(\mathbf{k}_X)$) the bounded derived category (resp. bounded from below derived category) of sheaves of $\mathbf{k}$-modules on X.

If M is a real manifold of class C^∞ and F a complex of sheaves, then the microsupport $\mathrm{SS}(F)$ of F is a closed conic (i.e. stable under $\mathbf{R}^+$-action) subset of the cotangent bundle of M. The precise definition is given in Section 3, but roughly speaking, $\mathrm{SS}(F)$ describes the (co)directions of X in which the cohomology of F cannot be extended. This notion can be interepreted by an analogy of Morse theory. Let φ be a real-valued function of class C^1. Then we can replace the notion of a critical point of φ with one with respect to F. That is to say, we can define a critical point with respect to F by the point p such that $d\varphi(p)$ does not belongs to $\mathrm{SS}(F)$. We can then obtain a analogous result of Morse theory in a framework of sheaf theory. The following is called (a special case of) the microlocal Morse lemma (cf. [GKS12, Proposition 1.8] or [KS90, Corollary 5.4.19]).

Proposition 1.1. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a function of class C^1, proper on $\mathrm{supp}(F)$. Let $a < b$ be in $\mathbf{R}$ and assume that $d\psi(x) \notin \mathrm{SS}(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, set $M_t := \psi^{-1}((-\infty, t))$. Then, the restriction morphism

$$\mathrm{R}\Gamma(M_b; F) \to \mathrm{R}\Gamma(M_a; F)$$

is isomorphism.

1.2 Truncated Microsupport

In 2003, Kashiwara, Monteiro-Fernandes, and Schapira introduced the notion of truncatd microsupprt in [KFS03], in course of the study of solutions of boundary value problems of $\mathcal{D}$-modules. The difference between microsupports and truncated microsupports is as follows. A point p in the cotangent bundle T^*M does not belong to the microsupport if and only if all the degree of a certain cohomology vanish, whereas p does not belong to the truncated microsupport if and only if the same kind of cohomology of all the degree under a fixed integer vanish. It was in the context of such kinds of analytic problems that they studied truncated microsupports.

1.3 Results

In this paper, in contrast to the previous research, we shall apply the truncated microsupport to study some geometrical problems. First we shall prove a truncated version of the noncharacteristic deformation lemma. This lemma is originally stated as follows.

Proposition 1.2 ([KS90, Proposition 2.7.2])**.** Let X be a Hausdorff space. Let $F \in \mathsf{D}^{+}(\mathbf{k}_X)$, and $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.
(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \mathrm{supp}\, F$ is compact.
(iii) Setting $Z_s := \bigcap_{t>s} \overline{U_t - U_s}$, we have for all pairs (s,t) with $s \leqq t$ and all $x \in Z_s - U_t$,

$$(\mathrm{R}\Gamma_{X-U_t}(F))_x = 0.$$

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$\mathrm{R}\Gamma\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} \mathrm{R}\Gamma\left(U_t; F\right).$$

This lemma is the most fundamental result in microlocal sheaf theory used all along [KS90]. Thus the truncated version of this lemma is also the key result in this paper. However, one should notice that the original proof of the lemma cannot be applied to our truncatd version. To prove this result, we weaken a hypothesis of the lemma and make an additional assumption of the family of deformation. Therefore, to make this assumption acceptable to the reader, we give several examples of families of deformation with figures. Then, in turn, we formulate truncated version of microlocal morse lemma, under the light of truncated microsupport. Note that by using the truncated microsupport, we can gain the information of the unique continuation. More precisely, if k is an integer and we consider the truncated microsupport $\mathrm{SS}_k(F)$ of a bounded complex of sheaves F, we can investigate whether the cohomology sheaves $H^j(F)$ propagate or not for $j < k$, and in particular, whether the continuation $H^k(F)$ is unique or not, if it can be extended. [*1]

1.4 Organization

This paper is organized as follows. In Section 2, we review the notations and results from [KS90]. In Section 4, we define truncated microsupport of sheaves, due to [KFS03]. In Section 4, we prove a truncated version of the noncharacteristic deformation lemma. Finally in Section 5, we apply the result in Section 4 to the microlocal Morse theory.

2 Background Material

In this section, we recall some definitions and results of homological algebra and sheaf theory including the theory of derived categories from [KS90], following its notations with slight modifications.

Let us denote by Z a locally closed set of X and by $i\colon Z \hookrightarrow X$ the inclusion. We use the following functors:

$$\mathrm{R}\Gamma_Z(\,\cdot\,) := \mathrm{R}i_*i^!, \quad (\,\cdot\,)_Z := \mathrm{R}i_!i^{-1}.$$

If there is no risk of confusion, we will write $\mathbf{k}_Z$ rather than $\mathbf{k}_{X,Z}$.

2.1 Geometrical Notions

Let M be a real manifold of class C^∞. We denote by $\pi_M\colon T^*M \to M$ the cotangent bundle of M. If there is no risk of confusion, we will write π rather than π_M. If $f\colon M \to N$ is a morphism of manifolds (i.e. a C^∞-map), then we have the natural diagram.

$$\begin{tikzcd}
TM \arrow[r, "f'"] \arrow[dr, "\tau_M"'] & M \underset{N}{\times} TN \arrow[r, "f_\tau"] \arrow[d] \arrow[dr, phantom, "\square"] & TN \arrow[d, "\tau_N"] \\
 & M \arrow[r, "f"'] & N,
\end{tikzcd}
\qquad
\begin{tikzcd}
T^*M \arrow[dr, "\pi_M"'] & M \underset{N}{\times} T^*N \arrow[l, "f_d"'] \arrow[r, "f_\pi"] \arrow[d] \arrow[dr, phantom, "\square"] & T^*N \arrow[d, "\pi_N"] \\
 & M \arrow[r, "f"'] & N.
\end{tikzcd}$$

[*1] この辺は拡張をいろいろな言い方で言い換えていますが，同じ方がいいですかね？

In this diagram, f_τ and f_π denote the natural projections, f' is defined by $(x;v) \mapsto (x; df_x v)$, and f_d is induced by the transposition of f'.

Approximation by Inductive Limits

Proposition 2.1 ([KS90, Proposition 2.5.1])**.** Let X be a topological space, Z a subspace, F a sheaf on X. Consider the canonical morhism:

$$\psi\colon \varinjlim_{U} \Gamma(U;F) \to \Gamma(Z;F),$$

where U ranges through the family of open neighborhoods of Z in X. Then the following hold.

(i) The morphism ψ is injective.
(ii) Assume that X is Hausdorff and Z is compact. Then ψ is an isomorphism.
(iii) Assume that X is paracompact and Z is closed. Then ψ is an isomorphism.

Note that a paracomact space is by definition a Hausdorff space.

2.2 Complement of Homological Algebra

Consider a projective system of abelian groups $(M_n, \rho_{n,p})_n$ indexed by $\mathbf{N}$. One says that this system satisfies the Mittag-Leffler condition (ML condition for short) if for any $n \in \mathbf{N}$, the decreasing sequence $(\rho_{n,p}(M_p))_p$ of subgroups of M_n is stationary.

For a projective system of abelian groups $(M_n)_n$, we set

$$M_\infty := \varprojlim_{n} M_n.$$

Consider a projective system of complexes

$$E_n^\bullet\colon \cdots \to M_n^{j-1} \to M_n^j \to M_n^{j+1} \to \cdots, \tag{2.1}$$

and its projective limit

$$E_\infty^\bullet\colon \cdots \to M_\infty^{j-1} \to M_\infty^j \to M_\infty^{j+1} \to \cdots. \tag{2.2}$$

Denote by

$$\Phi_k\colon H^k(E_\infty^\bullet) \to \varprojlim_{n} H^k(E_n^\bullet) \tag{2.3}$$

the natural morphism.

Proposition 2.2 ([KS90, Proposition 1.12.4])**.** Assume that for each $k \in \mathbf{Z}$, the system $\left(M_n^{k-1}\right)_n$ satisfies the ML condition. Then

(a) for each $k \in \mathbf{Z}$, the map Φ_k is surjective,
(b) if moreover, for a given i the system $\left(H^{i-1}(E_n^\bullet)\right)_n$ satisfies the ML condition, then Φ_i is bijective.

The following is called Kashiwara's lemma.

Proposition 2.3 ([KS90, Proposition 1.12.6])**.** Let $(X_s, \rho_{s,t} \colon X_t \to X_s)_{(s,(s \leqq t))}$ be a projective system of sets indexed by $\mathbf{R}$. Assume that for each $s \in \mathbf{R}$, the canonical maps

$$\lambda_s \colon X_s \to \varprojlim_{r<s} X_r$$

and

$$\mu_s \colon \varinjlim_{t>s} X_t \to X_s$$

are both injective (resp. surjective). Then all the maps ρ_{s_0,s_1} $(s_0 \leqq s_1)$ are injective (resp. surjective).

2.3 ML procedure for sheaves

Proposition 2.4 ([KS90, Proposition 2.7.1])**.** Let X be a topological space and let $F \in \mathsf{D}^+(\mathbf{Z}_X)$. Let $(U_n)_{n\in\mathbf{N}}$ be an increasing sequence of open subsets of X, $(Z_n)_{n\in\mathbf{N}}$ a decreasing sequence of closed subsets of X. Set $U := \bigcup_n U_n$, $Z := \bigcap_n Z_n$. Then the following hold.

(i) For any j, the natural map $\varphi_j \colon H^j_Z(U; F) \to \varprojlim_n H^j_{Z_n}(U_n; F)$ is surjective.

(ii) Assume that for a given j, the projective system $(H^{j-1}_{Z_n}(U_n; F))_n$ satisfies the ML condition. Then φ_j is bijective.

(iii) Let now $(X_n)_{n\in\mathbf{N}}$ be an increasing family of subsets of X satisfying $X = \bigcup_n X_n$ and $X_n \subset \mathrm{Int}(X_{n+1})$ for all n. Assume that for a given j, the projective system $\left(H^{j-1}(X_n; F)\right)_n$ satisfies the ML condition. Then the natural map $H^j(X; F) \to \varprojlim_n H^j(X_n; F)$ is bijective.

3 Truncated Microsupport

We give in this section the definition of a truncated microsupport and its functorial property under direct images of proper morphisms. Before that we review the original definition of a microsupport.

Definition 3.1. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}(F)$ of T^*X is defined by following condition: $p \notin \mathrm{SS}(F)$ if and only if there exists an open neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$\mathrm{R}\Gamma_{\{\varphi \geqq 0\}}(F)_x = 0. \tag{3.1}$$

Kashiwara, Monteiro-Fernandes, and Schapira modified this definition and gave the notion of a truncated microsupport.

Definition 3.2. Let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$. The closed conic subset $\mathrm{SS}_k(F)$ of T^*X is defined by: $p \notin \mathrm{SS}_k(F)$ if and only if there exists an open conic neighborhood U of p such that for any $x \in \pi(U)$ and for any real-valued C^1-function φ defined on a neighborhood of x such that $\varphi(x) = 0$, $d\varphi(x) \in U$, one has

$$H^j_{\{\varphi \geqq 0\}}(F)_x = 0 \quad \text{for any } j \leqq k. \tag{3.2}$$

Truncated microsupports satisfy the following functorial property under proper direct images.

Proposition 3.3. Let $f\colon X \to Y$ be a morphism of manifolds and let $F \in \mathsf{D}^{\mathrm{b}}(\mathbf{k}_X)$ such that f is proper on $\operatorname{supp}(F)$. Then for any $k \in \mathbf{Z}$,

$$\mathrm{SS}_k(\mathrm{R}f_*F) \subset f_\pi f_d^{-1}(\mathrm{SS}_k(F)). \tag{3.3}$$

The equality holds in case f is a closed embedding.

4 Truncated Noncharacteristic Deformation Lemma

In this section, we prove a truncated version of the noncharacteristic deformation lemma. First we state the result.

Theorem 4.1 (Truncated Noncharacteristic Deformation Lemma)**.** Let X be a Hausdorff space. Let $F \in \mathsf{D}^{+}(\mathbf{k}_X)$, and $(U_t)_{t\in\mathbf{R}}$ be a family of open subsets of X. Let $k \in \mathbf{Z}$. For $s \in \mathbf{R}$, set $Z_s \coloneqq \bigcap_{t>s} \overline{U_t - U_s}$. We assume the following conditions.

(i) $U_t = \bigcup_{s<t} U_s$, for all $t \in \mathbf{R}$.
(ii) For all pairs (s,t) with $s \leqq t$, the set $\overline{U_t - U_s} \cap \operatorname{supp} F$ is compact.
(iii) For all $s \in \mathbf{R}$ and all $x \in Z_s - U_s$, we have

$$(H^j_{X-U_s}(F))_x = 0 \quad \text{for all } j \leqq k.$$

(iv) For all pairs (s,t) with $s < t$, $Z_s \subset U_t$.

Then, for all $t \in \mathbf{R}$, we have the isomorphisms

$$H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \xrightarrow{\sim} H^j\left(U_t; F\right) \quad \text{for all } j < k,$$

and the injections

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^k\left(U_t; F\right).$$

Before we prove Theorem 4.1, let us give several examples of families of deformation.

Example 4.2. The condition (iv) of Theorem 4.1 is not contained in the original version (Proposition 1.2). Therefore let us give an example of $(U_t)_t$ that does not satisfy the condition (iv).

(i) Let $X = \mathbf{R}^2$, and for a real number t define the function

$$\varphi_t(x) \coloneqq \begin{cases} \max\left(0, t(2-|x|)\right) & (|x| < 2), \\ 0 & (|x| \geqq 2). \end{cases}$$

Then let us denote by U_t the open subset of X

$$U_t \coloneqq \left\{(x,y) \in X;\ y < \varphi_t(x)\right\}.$$

This family $(U_t)_t$ does not satisfy the condition (iv). Indeed, if $s < t$, we have

$$Z_s = \{(x, y) \in X;\ y = \varphi_s(x)\}$$

and the point $(2, 0) \in Z_s$ is not contained in U_t. Moreover, let us define a locally closed subset Z of X by

$$Z \coloneqq \left\{(x, y) \in X;\ (x+2)^2 + (y-1)^2 = 1\right\} \setminus \{(-1, 0)\}\,.$$

For this Z we denote by $\mathbf{k}_Z$ the sheaf on X as Section 2. If $j \leqq 0$, we have $H^j_{X \smallsetminus U_t}(\mathbf{k}_Z)_x = 0$ for all $t > s$ and $x \in Z_s \smallsetminus U_t$, but the morhism $H^0(U_1; \mathbf{k}_Z) = \mathbf{k} \to 0 = H^0(U_0; \mathbf{k}_Z)$ is not injective. This give an example that satisfies the conditions (i)–(iii) but not (iv), which results in that the consequence of the proposition does not hold. Hence the condition (iv) is necessary.

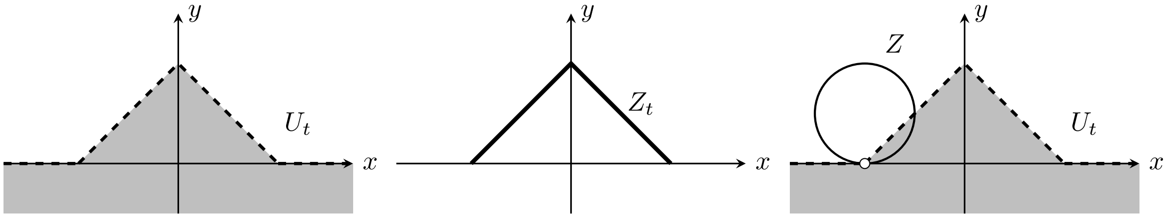

Fig.1　example (i)

Let us, in turn, give three examples of $(U_t)_t$ satisfying the condition (iv). In all the examples below, X is $\mathbf{R}^2$.

(ii) For a real number t, define the open set U_t of X by

$$U_t \coloneqq \begin{cases} \{(x, y) \in X;\ \|(x, y)\| < t\} & (t > 0), \\ \varnothing & (t \leqq 0). \end{cases}$$

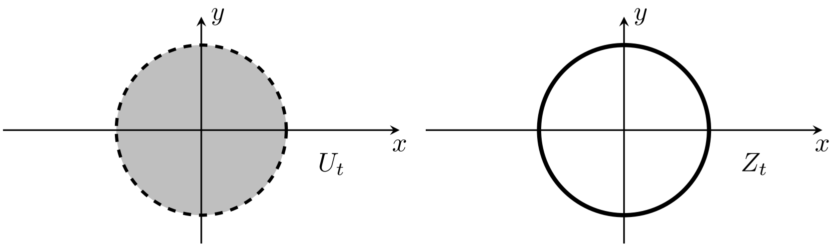

Fig.2　example (ii)

(iii) For a real number t, define the function $\varphi_t \colon \mathbf{R} \to \mathbf{R}$ as follows. If $t \leqq 0$, set $\varphi_t(x) \equiv 0$. If $t > 0$, set

$$\varphi_t(x) \coloneqq \begin{cases} \sqrt{t^2 - x^2} & (|x| < s), \\ 0 & (|x| \geqq s). \end{cases}$$

Then define the open set U_t of X by

$$U_t \coloneqq \{(x, y) \in X;\ y < \varphi_t(x)\}\,.$$

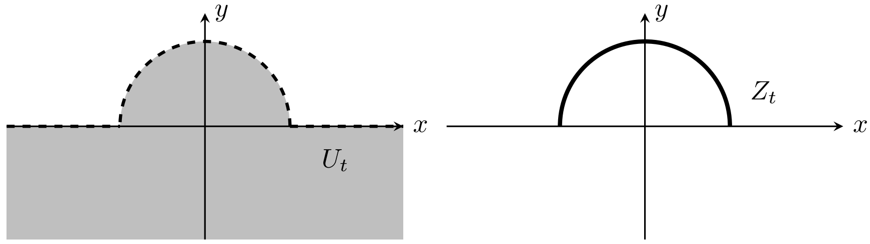

Fig.3 example (iii)

(iv) For a real number t, define the open set U_t of X as follows. If $t \leqq 0$, set

$$U_t := \{(x, y) \in X;\ y < 0\}.$$

If $t > 0$, set

$$U_t := \left\{ (x, y) \in X;\ \begin{array}{ll} \text{If } |x| < \frac{1}{t}, & y < t. \\ \text{Otherwise}, & y < \frac{1}{|x|}. \end{array} \right\}.$$

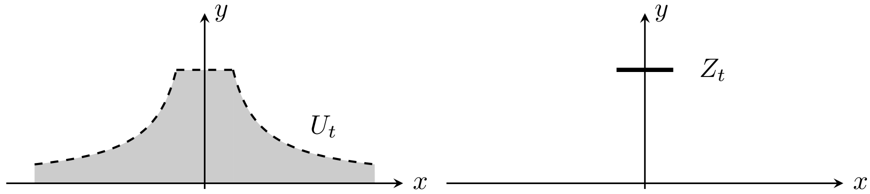

Fig.4 example (iv)

Proof of Theorem 4.1 Consider the following conditions for $j \in \mathbf{Z}$ and $s \in \mathbf{R}$:

$(a)^j_s$ $$\mu^j_s \colon \varinjlim_{t>s} H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F),$$

$(a')^k_s$ $$\mu^k_s \colon \varinjlim_{t>s} H^k(U_t; F) \hookrightarrow H^k(U_s; F),$$

$(b)^j_s$ $$\lambda^j_s \colon H^j(U_s; F) \xrightarrow{\sim} \varprojlim_{r<s} H^j(U_r; F).$$

Let us prove, for all $s \in \mathbf{R}$, $(a)^j_s$ for $j < k$ and $(a')^k_s$. We may assume that $\overline{U_t - U_s}$ is compact for any $s \leqq t$ by replacing X by $\operatorname{supp} F$. Consider the distinguished triangle

$$\mathrm{R}\Gamma_{X-U_s}(F) \to F \to \mathrm{R}\Gamma_{U_s}(F) \xrightarrow{+1}.$$

Let us endow with $Z_s \cup U_s$ the induced topology, and denote by $i_s \colon Z_s \cup U_s \hookrightarrow X$ the natural continuous map. Applying the functor

$$(\ \cdot\ )|_{Z_s \cup U_s} = i_s^{-1}$$

to the distiguished triangle above, we have

$$\mathrm{R}\Gamma_{X-U_s}(F)|_{Z_s\cup U_s} \to F|_{Z_s\cup U_s} \to \mathrm{R}\Gamma_{U_s}(F)|_{Z_s\cup U_s} \xrightarrow{+1}.$$

Again applying to this triangle the functor $\mathrm{R}\Gamma(Z_s \cup U_s;\ \cdot\ )$, we have

$$\mathrm{R}\Gamma(Z_s \cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)) \to \mathrm{R}\Gamma(Z_s \cup U_s; Z_s \cup U_s) \to \mathrm{R}\Gamma(U_s; F) \xrightarrow{+1} . \tag{4.1}$$

By the hypothesis (iii) and the definition of a local cohomology, we see that

$$H^j_{X-U_s}(F)\Big|_{Z_s \cup U_s} = 0 \quad \text{for any } j \leqq k,$$

and then

$$H^j\left(Z_s \cup U_s; \mathrm{R}\Gamma_{X-U_s}(F)\right) = 0 \quad \text{for any } j \leqq k.$$

Thus we have

$$H^j(Z_s \cup U_s; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for any } j < k, \tag{4.2}$$

$$H^k(Z_s \cup U_s; F) \hookrightarrow H^k(U_s; F). \tag{4.3}$$

Therefore, if we show Lemma 4.3 below, we have

$$\begin{aligned} \varinjlim_{t>s} H^j(U_t; F) &\cong H^j(Z_s \cup U_s; F) \\ &\xrightarrow{\sim} H^j(U_s; F) \end{aligned}$$

for any $j < k$, and $(a)^j_s$ follows. Similarly, we have

$$\begin{aligned} \varinjlim_{t>s} H^k(U_t; F) &\cong H^k(Z_s \cup U_s; F) \\ &\hookrightarrow H^k(U_s; F) \end{aligned}$$

and $(a')^k_s$ follows.

Lemma 4.3. For any $j \in \mathbf{Z}$ and $s \in \mathbf{R}$, there exists a natural isomorphism

$$\varinjlim_{t>s} H^j(U_t; F) \cong H^j(Z_s \cup U_s; F).$$

Proof of Lemma 4.3 First we prove that for $F \in \mathrm{Mod}(\mathbf{k}_X)$, the natural morphism

$$\varinjlim_{t>s} \Gamma(U_t; F) \xrightarrow{\sim} \Gamma(Z_s \cup U_s; F). \tag{4.4}$$

is isomorphic. Consider the following diagram.

$$\begin{tikzcd}
\varinjlim\limits_{t>s} \Gamma(U_t; F) \arrow[d, "\tilde{\phi}"'] \arrow[dr] & \\
\varinjlim\limits_{U \supset Z_s} \Gamma(U \cup U_s; F) \arrow[r, "\phi"'] \arrow[d] & \Gamma(Z_s \cup U_s; F|_{Z_s \cup U_s}) \arrow[d] \\
\varinjlim\limits_{U \supset Z_s} \Gamma(U; F) \arrow[r, "\psi"'] & \Gamma(Z_s; F|_{Z_s})
\end{tikzcd}$$

Let us begin by showing that the morphism

$$\tilde{\phi}\colon \varinjlim_{t>s} \Gamma(U_t;F) \to \varinjlim_{U\supset Z_s} \Gamma(U\cup U_s;F)$$

is an isomorphism. To prove this, it is enough to show that $(U_t)_{t>s}$ is a cofinal family of $(U\cup U_s)_{U\supset Z_s}$. Let U be an open neighborhood of Z_s. Then we shall show by contradiction that there exists a real number $t>s$ such that $U_t \subset U\cup U_s$. Let us assume that $U_t \setminus (U\cup U_s) \neq \varnothing$ for any $t>s$. Then define the family of subsets $(A_r)_{s<r<t}$ of $\overline{U_t\setminus U_s}$ by $A_r := U_r \setminus (U\cup U_s)$. This $(A_r)_{s<r<t}$ satisfies the finite intersection property. Indeed, for finite real numbers $s<r_0<\cdots<r_m<t$, it follows from the hypothesis that

$$\begin{aligned} A_{r_0}\cap\cdots\cap A_{r_m} &= A_{r_0} \\ &= U_{r_0}\setminus (U\cup U_s) \\ &\neq \varnothing. \end{aligned}$$

Therefore, since $\overline{U_t\setminus U_s}$ is compact, we have $\bigcap_{s<r<t}\overline{A_r}\neq\varnothing$. However, we have from the hypothesis $Z_s\subset U$ that

$$\begin{aligned} \varnothing \neq \bigcap_{s<r<t}\overline{A_r} &= \bigcap_{s<r<t}\overline{U_r\setminus(U\cup U_s)} \\ &= \bigcap_{s<r<t}(\overline{U_r\setminus U_s}\cap\overline{U_r\setminus U}) \\ &= \bigcap_{s<r<t}\overline{U_r\setminus U_s}\cap\bigcap_{s<r<t}\overline{U_r\setminus U} \\ &= Z_s\cap\bigcap_{s<r<t}\overline{U_r\setminus U} \\ &= Z_s\cap\bigcap_{s<r<t}\overline{U_r}\cap\overline{X\setminus U} \\ &= Z_s\cap\bigcap_{s<r<t}\overline{U_r}\cap(X\setminus U) \\ &= Z_s\cap(X\setminus U)\cap\bigcap_{s<r<t}\overline{U_r} \\ &= \varnothing. \end{aligned}$$

This is a contradiction, and it follows that $\tilde{\phi}$ is an isomorphism.

By this isomorphism, it is enough to show that

$$\phi\colon \varinjlim_{U\supset Z_s}\Gamma(U\cup U_s;F)\to\Gamma(Z_s\cup U_s;F)$$

is isomorphic. Since X is Hausdorff and Z_s is compact, the natural morhism

$$\psi\colon \varinjlim_{U\supset Z_s}\Gamma(U;F)\to\Gamma(Z_s;F)$$

is isomorphic by Proposition 2.1. Since U_s is open, we have another isomorphism

$$\psi'\colon \varinjlim_{V\supset U_s}\Gamma(V;F)\to\Gamma(U_s;F).$$

The injectivity of ϕ follows from the definition of a sheaf and the injectivities of ψ and ψ'.

Let us prove that ϕ is surjective. Let $\sigma \in \Gamma(Z_s; F)$ be a section of $F|_{Z_s \cup U_s}$. By the isomorphism ψ above, we can extend $\sigma|_{Z_s}$ to an open neighborhood U of Z_s in X. More precisely, there exists a section τ on U such that $\tau|_{Z_s} = \sigma|_{Z_s}$. Now we see that $\tau|_{Z_s} = \left(\tau|_{(Z_s \cup U_s) \cap U}\right)\Big|_{Z_s}$, and by the definition of a section on the closed subset Z_s of $(Z_s \cup U_s) \cap U$, we can find an open subset W of $(Z_s \cup U_s) \cap U$ such that

$$\tau|_W = \sigma|_W. \tag{4.5}$$

By the definition of the induced topology of $(Z_s \cup U_s) \cap U$, there exists an open set $\widetilde{W}$ of X such that

$$W = ((Z_s \cup U_s) \cap U) \cap \widetilde{W} \quad \text{and} \quad \widetilde{W} \subset U.$$

Thus we have $W = ((Z_s \cup U_s)) \cap \widetilde{W}$. For this $\widetilde{W}$, the pair $\left(Z_s \cup U_s, \widetilde{W}\right)$ is an open covering of $\widetilde{W} \cup U_s$. Thus by (4.5), we can glue σ and $\tau|_{\widetilde{W}}$ together and get a section $\widetilde{\sigma}$ such that

$$\widetilde{\sigma}|_{Z_s \cup U_s} = \sigma, \quad \widetilde{\sigma}|_{\widetilde{W}} = \tau|_{\widetilde{W}}.$$

This proves the surjectivity of ϕ.

Next Let $F \in \mathsf{D}^+(\mathbf{k}_X)$. Let $I^\bullet$ be a complex of flabby sheaves quasi-isomorphic to F. Then it follows from (4.4) that

$$\begin{aligned}
\varinjlim_{t>s} H^j(U_t; F) &\cong \varinjlim_{t>s} H^j\Gamma(U_t; I^\bullet) \\
&\cong H^j\left(\varinjlim_{t>s} \Gamma(U_t; I^\bullet)\right) \\
&\cong H^j\Gamma(Z_s \cup U_s; I^\bullet) \\
&\cong H^j(Z_s \cup U_s; F).
\end{aligned}$$

This completes the proof. □

Let us come back to the proof of Theorem 4.1. We prove $(b)^j_s$ for all $j \in \mathbf{Z}$ and $s \in \mathbf{R}$ by induction on j. Since $F \in \mathsf{D}^+(\mathbf{k}_X)$, we have $(b)^j_s$ for $j \ll 0$ and $s \in \mathbf{R}$. Hence there exists an integer $j_0 < k$ such that $(b)^j_s$ holds for any $j < j_0$ and $s \in \mathbf{R}$. Then assume $(b)^j_s$ is proved for all $j < j_0$ and $s \in \mathbf{R}$. Since we have $(a)^j_s$ and $(b)^j_s$ for each $j < j_0$ and each $s \in \mathbf{R}$, we can apply Kashiwara's lemma (Proposition 2.3) and have

$$H^j(U_t; F) \xrightarrow{\sim} H^j(U_s; F) \quad \text{for } j < j_0 \text{ and } s \leqq t.$$

Fix $s \in \mathbf{R}$ and consider the projective system $\left(H^{j_0-1}(U_{s-1/n})\right)_{n\in\mathbf{N}}$. This system satisfies ML condition. Therefore, by applying Proposition 2.4 (iii), we have

$$H^j(U_s; F) \xrightarrow{\sim} \varprojlim_n H^j\left(U_{s-\frac{1}{n}}; F\right).$$

Since $(s - 1/n)_n$ is a cofinal family of $(t)_{s>t}$, we have $(b)^{j_0}_s$. Then by induction on j, we have $(b)^j_s$ for

$j < k$ and $s \in \mathbf{R}$. Again by applying Proposition 2.4 to the projective system $\left(H^j(U_n; F)\right)_n$, we obtain

$$\begin{aligned} H^j\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) &\cong H^j\left(\bigcup_{n\in\mathbf{N}} U_n; F\right) \\ &\xrightarrow{\sim} \varprojlim_n H^j\left(U_n; F\right) \end{aligned}$$

and for each n, we have by Kashiwara's lemma (Proposition 2.3) that

$$H^j\left(U_n; F\right) \cong H^j(U_s; F)$$

for $n \geqq s$.

Finally we replace $(a)^j_s$ by $(a')^k_s$ for $j = k$, and obtain

$$H^k\left(\bigcup_{s\in\mathbf{R}} U_s; F\right) \hookrightarrow H^j\left(U_s; F\right).$$

This completes the proof. □

5 An Application to Microlocal Morse Theory

In this section, we shall apply the result in the previous section to a particular case of a truncated version of microlocal Morse lemma.

Let M be a manifold of class C^∞. If $t \in \mathbf{R}$, we denote the open interval $(-\infty, t)$ by I_t.

Lemma 5.1. Let $-\infty < a < b < +\infty$ be real numbers and $G \in \mathsf{D}^{\mathrm{b}}(\mathbf{R})$. Let k be an integer. Assume that

$$\left(H^j_{[t,+\infty)}(G)\right)_t \cong 0 \quad \text{for all } t \in [a, b)\ , \text{ and all } j \leqq k.$$

Then one has

$$H^j((-\infty, b)\,; G) \xrightarrow{\sim} H^j((-\infty, a)\,; G)$$

for all $j < k$, and this morphism is injective for $j = k$.

Proof. We will use Theorem 4.1. Let us put $X = (-\infty, b)$, $F = G|_{I_b}$, and $U_t = I_t$, and verify the hypotheses of the proposition.

(i) $\bigcup_{s<t} I_s = \bigcup_{s<t}(-\infty, s) = (-\infty, t) = I_t$ for all $t \in \mathbf{R}$.

(ii) If $s \leqq t$, then $\overline{I_t - I_s} \cap \operatorname{supp} G = \overline{[s, t)} \cap \operatorname{supp} G = [s, t] \cap \operatorname{supp} G$ is compact.

(iii) For $s \in \mathbf{R}$, we have

$$Z_s = \bigcap_{t>s} \overline{I_t - I_s} = \bigcap_{t>s} \overline{[s, t)} = \bigcap_{t>s} [s, t] = \{s\}\,,$$

and

$$Z_s - I_s = \{s\}\,.$$

By the hypothesis, we have

$$H^j_{X-I_s}(F)_s = H^j_{[s,\infty)}(F)_s = 0$$

for $s \in \{s\}$.

(iv) It follows that $Z_s = \{s\} \subset (-\infty, t) = I_t$ if $s < t$.

Then we can apply Theorem 4.1, and we have the isomorphisms for all $t \in [a, b)$ and $j < k$,

$$H^j(I_b; F) = H^j\left(\bigcup_s I_s; F\right) \xrightarrow{\sim} H^j(I_t; F)$$

and the injections

$$H^k(I_b; F) = H^k\left(\bigcup_s I_s; F\right) \hookrightarrow H^k(I_t; F).$$

□

Theorem 5.2. Let F in $\mathsf{D}^{\mathrm{b}}(\mathbf{k}_M)$, and $\psi\colon M \to \mathbf{R}$ be a real-valued function of class C^1, proper on $\operatorname{supp}(F)$. Let $a < b$ be real numbers. We assume $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$. For $t \in \mathbf{R}$, we set $M_t := \psi^{-1}((-\infty, t))$. Then, the restriction morphism

$$H^j(M_b; F) \to H^j(M_a; F)$$

is isomorphic for each $j < k$, and injective for $j = k$.

Proof. First, we have

$$H^j(M_t; F) \cong H^j\,\mathrm{R}\Gamma((-\infty, t)\,; \mathrm{R}\psi_* F)$$

for $t \in I$. Since we assume that $d\psi(x) \notin \mathrm{SS}_k(F)$ for $a \leqq \psi(x) < b$, and by (3.3), the formula $\mathrm{SS}_k(\mathrm{R}\psi_* F) \subset \psi_\pi \psi_d^{-1}\,\mathrm{SS}_k(F)$ holds. Hence we have

$$\left(H^j_{[\psi(x), +\infty)}(\mathrm{R}\psi_* F)\right)_{\psi(x)} = 0$$

for each $j \leqq k$. Then we can apply Lemma 5.1 and have

$$H^j((-\infty, b)\,; \mathrm{R}\psi_* F) \xrightarrow{\sim} H^j((-\infty, a)\,; \mathrm{R}\psi_* F)$$

for all $j < k$ and

$$H^k((-\infty, b)\,; \mathrm{R}\psi_* F) \hookrightarrow H^k((-\infty, a)\,; \mathrm{R}\psi_* F).$$

Then we can conclude that

$$H^j(M_b\,; F) \xrightarrow{\sim} H^j(M_a\,; F)$$

for all $j < k$ and

$$H^k(M_b\,; F) \hookrightarrow H^k(M_a\,; F).$$

□

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